

Midterm Exercise Set

Vectors, motion, and partial derivatives

Name: _____

Date: _____

56 problems with working space

Work in exact form unless a numerical approximation is requested. Show the setup and reasoning, label sketches, and include units in applications. This is a revision collection, not a single timed examination. No solutions are printed in this booklet.

Coverage

Problems	Topics	Pages
1–8	Coordinates, vectors, dot and cross products, lines and planes; Thomas 11.1–11.5	2–5
9–12	Vector functions, integration, projectile motion, arc length; Thomas 12.1–12.3	6–7
13–28	Domains, limits, partial derivatives, chain rule, gradients, tangent planes and differentials; Thomas 13.1–13.6	8–15
29–36	Local and absolute extrema, Lagrange multipliers; Thomas 13.7–13.8	16–23
37–40	Chapter 13 extension: Taylor expansions and constrained partial derivatives; Thomas 13.9–13.10	24–25
41–48	L1/L2 lesson-note supplement: differentials, chain rule, tangent lines, approximation and optimization	26–30
49–56	Selected ExerciseSet problems: motion, mixed partials, geometry, coordinate changes and optimization	31–36

Scope and sources

The core follows the requested textbook sections and reaches Lagrange multipliers, the topic of MIT 18.02 Lecture 13 in the supplied Fall 2007 notes. Problems 37–40 complete the requested Chapter 13 coverage beyond that lecture cutoff. Problem 46 also uses a Taylor polynomial.

Problems 1–40 are selected or adapted from the supplied *Thomas' Calculus: Early Transcendentals*, 15th edition in SI units, except Problem 35, from the MIT Lecture 13 notes. Problems 41–48 use the supplied L1/L2 question sheets. Each problem identifies its source exercise or PDF page. Instructions are restated; mathematical data are retained. Added follow-up tasks are marked.

Problems 49–56 are selected from the supplied `MA1521_Midterm_ExerciseSet.tex`, cited below as *ExerciseSet*. Its original problem numbers and source attributions are retained in the reference lines.

EXERCISES 11.1–11.2

Coordinates and vectors

1. A sphere in space

Thomas 11.1, Exercise 59

Find the center and radius of

$$x^2 + y^2 + z^2 - 4x + 6y - 10z = 11.$$

Write the equation in standard form.

WORKING

2. Displacement and direction

Thomas 11.2, Exercises 17 and 33; added unit-vector task

For $P = (5, 7, -1)$ and $Q = (2, 9, -2)$, find $\overrightarrow{PQ}$, its length, and its unit direction vector. Separately, find a vector of length 7 in the direction of $\mathbf{v} = 12\mathbf{i} - 5\mathbf{k}$.

WORKING

EXERCISES 11.3–11.4

Products and geometry

3. Dot product and projection

Thomas 11.3, Exercise 3

Let $\mathbf{v} = \langle 10, 11, -2 \rangle$ and $\mathbf{u} = \langle 0, 3, 4 \rangle$. Find $\mathbf{v} \cdot \mathbf{u}$, both lengths, the cosine of the angle between them, the scalar component of $\mathbf{u}$ along $\mathbf{v}$, and $\text{proj}_{\mathbf{v}} \mathbf{u}$.

WORKING

4. Triangle in space

Thomas 11.4, Exercise 15

Given $P = (1, -1, 2)$, $Q = (2, 0, -1)$, and $R = (0, 2, 1)$, find the area of triangle PQR and a unit vector perpendicular to its plane.

WORKING

EXERCISES 11.4–11.5

Volumes, lines, and planes

5. Scalar triple product

Thomas 11.4, Exercise 20

Let $\mathbf{u} = \langle 1, -1, 1 \rangle$, $\mathbf{v} = \langle 2, 1, -2 \rangle$, and $\mathbf{w} = \langle -1, 2, -1 \rangle$. Verify the cyclic identity

$$(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w} = (\mathbf{v} \times \mathbf{w}) \cdot \mathbf{u} = (\mathbf{w} \times \mathbf{u}) \cdot \mathbf{v}.$$

Find the volume of the parallelepiped determined by these vectors.

WORKING

6. An intersection and its plane

Thomas 11.5, Exercise 27

Find the intersection of the lines

$$\ell_1 : (x, y, z) = (2t + 1, 3t + 2, 4t + 3), \quad \ell_2 : (x, y, z) = (s + 2, 2s + 4, -4s - 1).$$

Then find the equation of the plane containing both lines.

WORKING

EXERCISES 11.5

Distances and perpendicularity

7. Distance to a line

Thomas 11.5, Exercise 34

Find the shortest distance from the origin to

$$x = 5 + 3t, \quad y = 5 + 4t, \quad z = -3 - 5t.$$

WORKING

8. Plane and point-to-plane distance

Thomas 11.5, Exercises 25 and 39

(a) Find the plane through $(2, 4, 5)$ perpendicular to the line $x = 5 + t, y = 1 + 3t, z = 4t$.**(b)** Find the distance from $(2, -3, 4)$ to the plane $x + 2y + 2z = 13$.

WORKING

EXERCISES 12.1–12.2

Vector functions and integration

9. Motion and a tangent line

Thomas 12.1, Exercise 23; added velocity and acceleration tasks

For

$$\mathbf{r}(t) = \langle \sin t, t^2 - \cos t, e^t \rangle,$$

find the velocity and acceleration vectors. Find parametric equations for the tangent line at $t = 0$.

WORKING

10. Recovering position

Thomas 12.2, Exercise 14

Solve the vector initial-value problem

$$\mathbf{r}'(t) = \langle t^3 + 4t, t, 2t^2 \rangle, \quad \mathbf{r}(0) = \langle 1, 1, 0 \rangle.$$

WORKING

EXERCISES 12.2–12.3

Projectile motion and arc length

11. An ideal projectile

Thomas 12.2, Exercise 25

A projectile starts at ground level with speed 500 m s^{-1} at an angle of 45° above the horizontal. Neglect air resistance and use $g = 9.8 \text{ m s}^{-2}$. Find its flight time, horizontal range, height when it is 5 km downrange, and maximum height.

WORKING

12. Unit tangent and arc length

Thomas 12.3, Exercises 2 and 11

(a) Find the unit tangent and length of $\mathbf{r}(t) = \langle 6 \sin 2t, 6 \cos 2t, 5t \rangle$, $0 \leq t \leq \pi$.

(b) For $\mathbf{r}(t) = \langle 4 \cos t, 4 \sin t, 3t \rangle$, find $s(t)$ measured from $t = 0$ and the length for $0 \leq t \leq \pi/2$.

WORKING

EXERCISES 13.1

Domains and level curves

13. Domains with boundaries

Thomas 13.1, Exercises 6 and 10

Find and sketch the domain of each function. Indicate whether each boundary is included.

$$(a) \quad f(x, y) = \ln(x^2 + y^2 - 4), \quad (b) \quad g(x, y) = \ln(xy + x - y - 1).$$

WORKING

14. Reading a function through its level curves

Thomas 13.1, Exercises 19 and 25; level values specified

For each function, find its domain and range, and sketch the level curves for the stated values of c :

$$(a) \quad f(x, y) = 4x^2 + 9y^2, \quad c = 0, 1, 4,$$

$$(b) \quad g(x, y) = \ln(x^2 + y^2), \quad c = -1, 0, 1.$$

WORKING

EXERCISES 13.2

Limits and continuity

15. A limit that exists

Thomas 13.2, Exercise 21; added continuity task

Evaluate

$$\lim_{(x,y) \rightarrow (0,0)} \frac{\sin(x^2 + y^2)}{x^2 + y^2}.$$

Justify your answer for arbitrary approaches to the origin. How should the function be defined at the origin to make it continuous?

WORKING

16. A limit that fails

Thomas 13.2, Exercise 42; added curved-path comparison

Show that

$$f(x, y) = \frac{x^4}{x^4 + y^2}$$

has no limit at the origin. Also compare its limits along $y = mx$ for $m \neq 0$ with its limits along $y = kx^2$. Explain why checking a selection of straight lines cannot prove existence of a two-variable limit.

WORKING

EXERCISES 13.3

Partial differentiation

17. First and second partial derivatives

Thomas 13.3, Exercises 16 and 42

- (a) Find f_x and f_y for $f(x, y) = e^{xy} \ln y$, $y > 0$.
- (b) For $g(x, y) = \sin(xy)$, find all four second-order partial derivatives and verify $g_{xy} = g_{yx}$.

WORKING

18. Definition and implicit differentiation

Thomas 13.3, Exercises 63 and 69

- (a) Use the limit definition to find $f_x(1, 2)$ and $f_y(1, 2)$ for $f(x, y) = 1 - x + y - 3x^2y$.
- (b) The equation $xy + xz^3 - 2yz = 0$ defines z locally as a function of x, y near $(1, 1, 1)$. Find z_x there, with y held fixed.

WORKING

EXERCISES **13.4**

The chain rule

19. One independent variable

Thomas 13.4, Exercise 4

Let $w = \ln(x^2 + y^2 + z^2)$, where $x = \cos t$, $y = \sin t$, and $z = 4t$. Find dw/dt using the chain rule, then verify it by direct substitution. Evaluate the derivative at $t = 3$.

WORKING

20. Two independent variables

Thomas 13.4, Exercise 9

Let

$$w = xy + yz + xz, \quad x = u + v, \quad y = u - v, \quad z = uv.$$

Find w_u and w_v by the chain rule and by direct substitution. Evaluate both at $(u, v) = (1/2, 1)$.

WORKING

EXERCISES 13.4–13.5

Implicit surfaces and gradients

21. Implicit partial derivatives

Thomas 13.4, Exercise 31; added regularity check

For

$$z^3 - xy + yz + y^3 - 2 = 0,$$

find z_x and z_y at $(1, 1, 1)$. Check the nonzero partial derivative that permits solving locally for z as a differentiable function of x, y .

WORKING

22. Gradient and a level curve

Thomas 13.5, Exercise 2; added tangent-line task

For $f(x, y) = \ln(x^2 + y^2)$, calculate $\nabla f(1, 1)$. Sketch the level curve through $(1, 1)$ and the gradient vector based there. Find the tangent line to this level curve at $(1, 1)$.

WORKING

EXERCISES 13.5

Directional derivatives

23. A direction in three dimensions

Thomas 13.5, Exercise 15

Find the directional derivative of

$$f(x, y, z) = xy + yz + zx$$

at $P = (1, -1, 2)$ in the direction of $\mathbf{v} = \langle 3, 6, -2 \rangle$. Show the unit vector used in your calculation.

WORKING

24. Largest, smallest, and zero rates

Thomas 13.5, Exercise 29(a)–(d)

Let $f(x, y) = x^2 - xy + y^2 - y$. At $(1, -1)$, find all unit directions $\mathbf{u}$ for which $D_{\mathbf{u}}f$ is largest, smallest, zero, or equal to 4. Give the derivative values in the first two cases.

WORKING

EXERCISES **13.6**

Tangent planes and intersecting surfaces

25. Tangent plane and normal line

Thomas 13.6, Exercise 4

Find the tangent plane and normal line to

$$x^2 + 2xy - y^2 + z^2 = 7$$

at $(1, -1, 3)$.

WORKING

26. Tangent to an intersection curve

Thomas 13.6, Exercise 16

Find parametric equations for the tangent line at $(1, 1, 1)$ to the intersection of

$$xyz = 1, \quad x^2 + 2y^2 + 3z^2 = 6.$$

WORKING

EXERCISES 13.6

Linearization and rates of change

27. Linearization with an error bound

Thomas 13.6, Exercise 35

Find the linearization of $f(x, y) = x^2 - 3xy + 5$ at $(2, 1)$. Find an upper bound for $|f(x, y) - L(x, y)|$ throughout

$$|x - 2| \leq 0.1, \quad |y - 1| \leq 0.1.$$

You may use either the exact remainder or a second-derivative bound.

WORKING

28. Temperature along a space curve

Thomas 13.6, Exercise 26

The temperature is $T(x, y, z) = 2x^2 - xyz$ degrees Celsius. A particle follows $x = 2t^2$, $y = 3t$, $z = -t^2$, with distance in meters and time in seconds. At $(8, 6, -4)$, find the rate of temperature change (a) per meter traveled and (b) per second.

WORKING

EXERCISES **13.7**

Local extrema

29. Classifying all critical points

Thomas 13.7, Exercises 14 and 19

For each function, find every critical point and classify it as a local maximum, local minimum, or saddle point. Give the function value at each point.

$$(a) \quad f(x, y) = x^3 + 3xy + y^3, \quad (b) \quad g(x, y) = 4xy - x^4 - y^4.$$

WORKING

EXERCISES **13.7**

When the second-derivative test fails

30. A parameter-dependent critical point

Thomas 13.7, Exercises 45–46; added borderline classification

Consider $f(x, y) = x^2 + kxy + y^2$, where $k \in \mathbb{R}$.

- (a) Show that $(0, 0)$ is always a critical point.
- (b) Determine the values of k for which the second-derivative test proves a local minimum or a saddle point.
- (c) When the test is inconclusive, classify the origin directly. State whether the extremum is strict, and find any other critical points.

WORKING

EXERCISES **13.7**

Absolute extrema on a closed region

31. A triangular domain

Thomas 13.7, Exercise 31

Find the absolute maximum and minimum of

$$f(x, y) = 2x^2 - 4x + y^2 - 4y + 1$$

on the closed triangle bounded by $x = 0$, $y = 2$, and $y = 2x$ in the first quadrant. Check the interior, all three edges, and the vertices. State where each extreme value occurs.

WORKING

EXERCISES **13.7**

Absolute extrema on a disk

32. Hottest and coldest points

Thomas 13.7, Exercise 41

A circular plate occupies $x^2 + y^2 \leq 1$, including its boundary. Its temperature is

$$T(x, y) = x^2 + 2y^2 - x.$$

Find the hottest and coldest temperatures and all points where they occur.

WORKING

EXERCISES **13.8**

Lagrange multipliers in the plane

33. Product on an ellipse

Thomas 13.8, Exercise 1

Use Lagrange multipliers to find the extreme values of $f(x, y) = xy$ on $x^2 + 2y^2 = 1$, and all points attaining them. Explain why your candidates include the absolute extrema.

WORKING

EXERCISES **13.8**

Lagrange multipliers in space

34. Linear function on a sphere

Thomas 13.8, Exercise 23; added geometric verification

Find the maximum and minimum of

$$f(x, y, z) = x - 2y + 5z \quad \text{on} \quad x^2 + y^2 + z^2 = 30.$$

Use Lagrange multipliers and give all points attaining the extreme values. Verify your answers using a dot-product inequality or a geometric argument.

WORKING

EXERCISES **13.8**[MIT Lecture 13 connection](#)**35. Closest points on a hyperbola**

MIT 18.02, Fall 2007, Lecture 13 example; added global-extrema task

Find the points on $xy = 3$ closest to the origin. Use Lagrange multipliers on the squared distance, retaining both branches of the constraint. Justify that the candidates are global minima. Does a farthest point exist? Explain.

WORKING

EXERCISES **13.8**

An optimization model

36. Largest box under a plane

Thomas 13.8, Exercise 28

A rectangular box lies in the first octant, with three faces in the coordinate planes and its opposite vertex on

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1, \quad a, b, c > 0.$$

Find the dimensions and volume of the largest such box. Set up the constraint and multiplier equations, and account for the boundary of the feasible region.

WORKING

EXERCISES **13.9**

Extension beyond MIT Lecture 13

37. Taylor polynomials

Thomas 13.9, Exercise 2

Use Taylor's formula at $(0, 0)$ to find the quadratic and cubic approximations of $f(x, y) = e^x \cos y$. Collect terms by total degree.

WORKING

38. A mixed expansion

Thomas 13.9, Exercise 5

Find the quadratic and cubic Taylor polynomials at $(0, 0)$ for

$$f(x, y) = e^x \ln(1 + y).$$

WORKING

EXERCISES 13.10

Extension beyond MIT Lecture 13

39. What is held fixed?

Thomas 13.10, Exercise 1

Let $w = x^2 + y^2 + z^2$, with $z = x^2 + y^2$. Draw the variable-dependence diagrams, then calculate

$$\left(\frac{\partial w}{\partial y}\right)_z, \quad \left(\frac{\partial w}{\partial z}\right)_x, \quad \left(\frac{\partial w}{\partial z}\right)_y.$$

Work locally where the chosen independent variables are valid.

WORKING

40. Constrained derivatives of internal energy

Thomas 13.10, Exercise 3

Let $U = f(P, V, T)$ and impose $PV = nRT$, where $n, R > 0$ are constants. Express

$$\left(\frac{\partial U}{\partial P}\right)_V \quad \text{and} \quad \left(\frac{\partial U}{\partial T}\right)_V$$

in terms of the formal partial derivatives f_P, f_V, f_T . Draw a dependence diagram for each choice of independent variables.

WORKING

EXERCISES 13.6

L1 supplement: total differentials

41. Explicit differentials

L1 Week 05 Mon question sheet, PDF page 2(a)–(c)

Find and simplify the total differential of each function:

$$(a) \quad w = \ln(xyz), \quad (b) \quad w = x^3 y^2 z, \quad (c) \quad z = \frac{x - y}{x + y}.$$

WORKING

42. Implicit differentials

L1 Week 05 Mon question sheet, PDF page 3(a)–(b)

Take differentials of both sides and solve algebraically for dw :

$$(a) \quad \frac{1}{w} = \frac{1}{t} + \frac{1}{u} + \frac{1}{v}, \quad (b) \quad u^2 + 2v^2 + 3w^2 = 10.$$

Treat w as a differentiable function of the other variables, locally where the displayed relation permits this.

WORKING

EXERCISES 13.4–13.6

L1/L2 supplement: chain rule and geometry

43. The chain rule by two methods

L1 Week 05 Mon question sheet, PDF page 4(a)–(c)

Find dw/dt first by the chain rule and then by substituting to express w directly in terms of t :

- (a) $w = xyz$, with $x = t$, $y = t^2$, $z = t^3$.
(b) $w = x^2 - y^2$, with $x = \cos t$, $y = \sin t$.
(c) $w = \ln(u^2 + v^2)$, with $u = 2 \cos t$, $v = 2 \sin t$.

WORKING

44. An intersection curve

L2 Week 05 2nd (Fri) question sheet, Q10.2

Find parametric equations of the tangent line at $(1, 2, 1)$ to the curve of intersection of

$$x^3 + 2xy + yz = 7, \quad 3x^2 - yz = 1.$$

WORKING

EXERCISES 13.6 and 13.9

L2 supplement: linear and quadratic approximation

45. Estimating a change

L2 Week 06 1st (Tue) question sheet, Q10.7

For $f(x, y, z) = xz + \sin(xyz)$, use linear approximation to estimate the change in f when the point moves from

$$\left(\frac{1}{2}, \pi, \frac{1}{3}\right) \quad \text{to} \quad \left(\frac{1}{2} + 0.03, \pi + 0.04, \frac{1}{3} - 0.02\right).$$

WORKING

46. A second-degree Taylor approximation

L2 Week 06 1st (Tue) question sheet, Q10.9; beyond MIT Lecture 13

Let $f(x, y) = e^{x+4y} \cos x$.

- (a) Find its second-degree Taylor polynomial at $(0, 0)$.
- (b) Use that polynomial to approximate $f(0.2, 0.1)$.

WORKING

EXERCISES 13.8

L2 supplement: an additional inequality

47. A circle and a restricted arc

L2 Week 06 1st (Tue) question sheet, Q11.1–Q11.2

Use Lagrange multipliers to maximize $f(x, y) = xy$:

- (a) Subject to $(x - 2)^2 + y^2 = 4$.
- (b) Subject to both $(x - 2)^2 + y^2 = 4$ and $x \leq 2$.

Give the maximizing points and values. In part (b), account for the endpoints created by the inequality as well as stationary points on the allowed arc.

WORKING

EXERCISES **13.8**

L2 supplement: minimum material cost

48. An open-top box

L2 Week 06 1st (Tue) question sheet, Q11.4

An open-top rectangular box must hold 6 m^3 . Material for the front and back faces costs \$10 per square meter, material for the two lateral faces costs \$5 per square meter, and the base costs \$30 per square meter. Use Lagrange multipliers to find the minimum material cost and the dimensions of the box. Define length x along the front edge, width y along a lateral edge, and height h , so the front and back faces each have area xh .

WORKING

EXERCISES 12.1 and 13.3

Selected problems: motion and mixed partials

49. Intersecting paths versus a collision

ExerciseSet, Problem 14; attributed there to L2 Q05.13

Two particles have position vectors

$$\mathbf{r}_1(t) = \langle t, t^2, t^3 \rangle, \quad \mathbf{r}_2(t) = \langle 1 + 2t, 1 + 6t, 1 + 14t \rangle.$$

- (a) Do the particles collide? Justify your answer.
- (b) Do their paths intersect? Justify your answer, using independent parameters for the two paths.

WORKING

50. Can these be the partial derivatives of a function?

ExerciseSet, Problem 20(b)–(c); attributed there to MIT PSet 3 and L2 Q07.6

In each case, use equality of mixed partial derivatives to determine the constant, then find a function $f(x, y)$ having the stated first partial derivatives.

- (a) $f_x = axy + 3y^2$, $f_y = x^2 + 6xy$.
- (b) $f_x = 3x^2 + Axy$, $f_y = 3x^2 + 6y^2$.

WORKING

EXERCISES 13.5–13.6

Selected problems: geometry and an unknown parameter

51. A tangent plane containing a ray

ExerciseSet, Problem 24; attributed there to MIT PSet 3

Let $P_0 = (x_0, y_0, z_0)$ lie on the cone $z = \sqrt{x^2 + y^2}$, with $z_0 > 0$.

- (a) Find the tangent plane to the cone at P_0 .
- (b) Parametrize the ray from the origin through P_0 . Show that the entire ray lies on both the cone and the tangent plane.

WORKING

52. Recovering a parameter from a directional derivative

ExerciseSet, Problem 35; attributed there to L2 Q09.9

Let $a > 0$ and

$$f(x, y) = (x^2 + a)e^{2x+3y^2}.$$

The directional derivative at $(-6, 2)$ in the direction of $\mathbf{i} + 2\mathbf{j}$ is $216\sqrt{5}$. Find a .

WORKING

EXERCISES 13.4

Selected problems: a change of coordinates

53. The chain rule in polar coordinates

ExerciseSet, Problem 30; attributed there to MIT PSet 4 and L2 Q09.6

Let f have continuous first and second partial derivatives, and define

$$w(r, \theta) = f(r \cos \theta, r \sin \theta), \quad r > 0.$$

All derivatives of f below are evaluated at $(r \cos \theta, r \sin \theta)$.

(a) Prove that

$$f_x^2 + f_y^2 = w_r^2 + \frac{1}{r^2} w_\theta^2.$$

(b) Prove that

$$w_{rr} = f_{xx} \cos^2 \theta + 2f_{xy} \sin \theta \cos \theta + f_{yy} \sin^2 \theta.$$

WORKING

EXERCISES **13.7**

Selected problems: least-squares fitting

54. Fitting a line and a quadratic

ExerciseSet, Problem 44; attributed there to MIT PSet 3 and L2 Q08.7–Q08.8

Use least squares, minimizing the sum of squared vertical residuals, to find:

- (a) The line $y = ax + b$ that best fits $(-1, 1)$, $(1, 2)$, and $(3, 2)$.
- (b) The quadratic of the form $y = ax^2 + bx$ that best fits $(-1, 1)$, $(1, 1)$, and $(2, 0)$.

Write the objective function and the equations determining its minimizing coefficients in each case.

WORKING

EXERCISES 13.8

Selected problems: two constraints

55. Extrema on intersections

ExerciseSet, Problem 50; attributed there to Thomas 13.8, Exercises 42–43

- (a) Find the extreme values of $f(x, y, z) = xy + z^2$ on the circle where $y - x = 0$ intersects $x^2 + y^2 + z^2 = 4$. Give all points attaining them.
- (b) Find the maximum of $w = xyz$ on the line where $x + y + z = 40$ intersects $x + y - z = 0$. Give a geometric or one-variable argument showing that the candidate is a maximum.

WORKING

EXERCISES **13.8**

Selected problems: interpreting multiplier candidates

56. Necessary does not mean sufficient

ExerciseSet, Problem 52; attributed there to Thomas 13.8, Exercise 45

Consider $f(x, y) = x + y$ subject to $xy = 16$.

- (a) Use Lagrange multipliers to find all stationary candidates on the constraint.
- (b) Explain why f has no absolute maximum on the whole hyperbola.
- (c) Classify each candidate as a constrained local maximum or minimum, specifying the branch on which it lies. Decide whether either is an absolute extremum on the whole constraint.

WORKING